NETWORK ENGINEERING

A complete, prerequisite-ordered learning roadmap

A practical roadmap from zero-to-foundation through enterprise networking, troubleshooting, automation, and ISP/service-provider concepts. It is organized so each module prepares you for the next; you should not need to stop halfway through a module to learn a major prerequisite.

Primary path	Networking fundamentals → CCNA-level skills → real-world operations → specialization
Core lab tools	Cisco Packet Tracer, Wireshark, Linux, Git/GitHub; later GNS3 or EVE-NG
Cost approach	Start with free software and virtual labs. Physical routers/switches are not required at the beginning.
Portfolio rule	Document commands, topology, expected result, actual result, failure, fix, and lessons learned.

How to use this document: Follow the module numbers in order. Do not rush because a topic looks familiar. Complete the lab and exit criteria before moving on. The module sequence is more important than a fixed calendar.

This roadmap is vendor-aware but not vendor-locked. It uses CCNA 200-301 as a useful syllabus spine, then adds Linux, monitoring, automation, documentation, and ISP concepts that make the learning more practical.

Roadmap map

The dependency chain below is the intended order. Later modules reuse earlier skills.

Stage	Modules	Focus
Foundation	1–4	Computer/network basics, command line, OSI/TCP-IP, Ethernet
Addressing	5–7	Binary, IPv4/subnetting, IPv6, core protocols
LAN switching	8–10	Switching, VLANs/trunks, STP, EtherChannel
Routing	11–13	Routing tables, static/default routes, OSPF, inter-VLAN routing
Network services & control	14–16	DHCP/DNS/NAT/ACLs, wireless, security basics
Operations	17–19	Troubleshooting, monitoring, documentation, Linux networking
Advanced practice	20–23	Virtual labs, automation, resilient design, ISP/service-provider topics
Career proof	24	Portfolio projects, interview practice, certification readiness

Rule: If a later lab exposes a gap, pause and review the relevant earlier module. That is targeted revision—not a reason to jump randomly between unrelated courses.

Module 1: Computer and Network Foundations

Goal: Understand what a network is and what devices actually do before memorizing configuration commands.

Learn in this order

- What a network is; LAN, WLAN, WAN, PAN, internet, intranet, extranet.
- Clients, servers, peers, endpoints, network interfaces, packets and frames (high-level meaning).
- Network devices: NIC, switch, router, wireless access point, modem/ONT, firewall, load balancer.
- Physical vs logical topology; star, mesh, point-to-point, hub-and-spoke.
- Bandwidth, throughput, latency, jitter, packet loss, availability; why they differ.
- Copper Ethernet, fiber, Wi-Fi, connectors, duplex, link speed, PoE; basic cabling awareness.
- Home network path: device → access point/switch → router → ISP → internet.

Hands-on lab / evidence

Draw your home network and label every device, interface, link type, and the likely path to a public website. Identify which device provides Wi-Fi, routing, and ISP handoff.

Move on when you can

You can explain the job of a switch versus a router, and trace a device's path from local network to the internet.

Module 2: Computer, OS, and Command-Line Basics

Prerequisites: Module 1

Goal: Build enough operating-system literacy to inspect interfaces and troubleshoot without relying only on GUI screens.

Learn in this order

- Files, folders, paths, permissions, processes, services, logs, environment variables.
- Windows terminal basics: cd, dir, ipconfig, ping, tracert, nslookup, route print, arp -a, netstat.
- Linux shell basics: pwd, ls, cd, cat, less, grep, ip, ping, traceroute, ss, dig, journalctl.
- Administrator/root privileges and why commands may fail without them.
- Install/uninstall software safely; read version information and official documentation.
- Basic text editing and saving command output to a lab log.

Hands-on lab / evidence

Capture output from ipconfig /all (Windows) or ip addr and ip route (Linux). Find your interface address, default gateway, DNS server, and route table.

Move on when you can

You can inspect an interface and route table from the command line and explain the key fields.

Module 3: Network Models and Encapsulation

Prerequisites: Modules 1–2

Goal: Learn the mental model used to reason about every protocol and troubleshoot layer by layer.

Learn in this order

- OSI 7-layer model and TCP/IP model; what each layer contributes.
- Encapsulation/decapsulation: application data, TCP/UDP segment, IP packet, Ethernet frame, bits.
- Addresses by layer: MAC address, IP address, TCP/UDP port, application name.
- Connection-oriented vs connectionless communication.
- Unicast, broadcast, multicast, anycast (conceptual distinctions).

- MTU, payload, overhead, fragmentation basics; why packet size matters.
- How to use a layered troubleshooting method without treating OSI as a rigid law.

Hands-on lab / evidence

Use Wireshark to capture a web request or ping. Identify Ethernet, IP, transport, and application details where visible.

Move on when you can

You can trace a sample packet through layers and explain which address changes at each stage.

Module 4: Ethernet, MAC, ARP, and Switching Fundamentals

Prerequisites: Module 3

Goal: Understand local-network delivery before configuring VLANs or routing.

Learn in this order

- Ethernet frame fields; source/destination MAC; EtherType; FCS at a conceptual level.
- MAC address format and unicast/multicast/broadcast MACs.
- How a switch learns its MAC address table and forwards/floods frames.
- Unknown unicast, broadcast domain, collision domain (historical/current context).
- ARP purpose, ARP request/reply, ARP cache, gratuitous ARP; IPv4 neighbor discovery.
- Duplex/speed negotiation, link lights, interface status and common physical faults.
- Basic switch CLI concepts: user/privileged/ global/interface configuration modes.

Hands-on lab / evidence

In Packet Tracer, connect two PCs through a switch. Inspect the MAC table, generate traffic, and observe ARP and frame delivery in Simulation mode.

Move on when you can

You can explain how a switch learns where a host is and why a host needs ARP before sending to a local IPv4 neighbor.

Module 5: Binary, Hexadecimal, and IPv4 Addressing

Prerequisites: Modules 1 and 3

Goal: Learn the number system and address structure required for subnetting.

Learn in this order

- Decimal, binary, hexadecimal; powers of two and bit positions.
- IPv4 address as 32 bits; network portion and host portion.
- Subnet mask and CIDR prefix notation (/8, /16, /24 etc.).
- Network address, usable host range, broadcast address, default gateway.
- Private IPv4 ranges, public IPv4, loopback, link-local/APIPA, unspecified and limited broadcast.
- Classful addressing as history only; modern classless CIDR.
- Address planning and avoiding duplicate IPs.

Hands-on lab / evidence

Convert several octets between decimal and binary. For a /24 and /26, identify network, broadcast, usable range, and host count.

Move on when you can

You can read an IPv4 address with a prefix and explain which bits identify the subnet.

Module 6: IPv4 Subnetting and Address Planning

Prerequisites: Module 5

Goal: Make subnetting reliable enough to perform quickly and correctly in network design.

Learn in this order

- Subnetting by borrowing host bits; block size/increment method.
- Calculate network ID, broadcast, first/last usable addresses, and total/usable hosts.
- Subnet a larger block into equal-sized subnets and into variable-sized subnets.
- VLSM: allocate address space to different host requirements.
- Route summarization/supernetting fundamentals and when aggregation is valid.
- Plan IP ranges for users, servers, management, voice, guests, and point-to-point links.
- Common subnetting mistakes: off-by-one errors, wrong mask, overlapping subnets.

Hands-on lab / evidence

Complete an address plan for a small office with several departments and a server segment. Include requirements, prefix, gateway, usable range, and rationale.

Move on when you can

You can solve subnetting problems accurately without guessing and can spot overlapping address ranges.

Module 7: IPv6 Addressing and Neighbor Discovery

Prerequisites: Modules 5–6

Goal: Understand modern IP addressing and avoid treating IPv6 as an optional mystery.

Learn in this order

- Why IPv6 exists; 128-bit addresses; hexadecimal notation and zero compression.
- Global unicast, unique local, link-local, loopback, unspecified, multicast, anycast.
- Prefix length and subnetting basics; /64 convention and exceptions.
- SLAAC, DHCPv6 stateful/stateless concepts, router advertisements.
- ICMPv6 and Neighbor Discovery; how IPv6 replaces ARP's role.
- Dual-stack, IPv4/IPv6 coexistence, DNS AAAA records, basic transition concepts.
- IPv6 default routes and basic address troubleshooting.

Hands-on lab / evidence

Configure IPv6 addresses on a small Packet Tracer topology and verify connectivity. Identify link-local and global addresses.

Move on when you can

You can read compressed IPv6 notation, recognize common address types, and explain SLAAC at a high level.

Module 8: Transport and Essential Network Protocols

Prerequisites: Modules 3, 5, and 6

Goal: Understand the protocols that make applications usable across networks.

Learn in this order

- TCP vs UDP; ports, sockets, three-way handshake, sequencing, acknowledgments, retransmission.
- Well-known service ports: DNS 53, DHCP 67/68, HTTP 80, HTTPS 443, SSH 22, SMTP 25, NTP 123, SNMP 161/162.
- ICMP/ICMPv6: echo, unreachable, time exceeded; ping and traceroute limitations.
- DNS resolution: recursive resolver, authoritative server, A/AAAA, CNAME, MX, PTR, TTL.

- DHCP DORA process, leases, scopes/pools, reservations, relay concept.
- HTTP/HTTPS and TLS purpose; certificate and name mismatch basics.
- NTP, syslog, SNMP purpose; what network devices need to operate and be monitored.

Hands-on lab / evidence

Use Wireshark to compare DNS lookup, ping, and a TCP connection. Use nslookup/dig and observe the DNS answer and TTL.

Move on when you can

You can distinguish transport from application protocols and explain what DNS, DHCP, ICMP, and TCP each contribute.

Module 9: Cisco CLI and Basic Device Configuration

Prerequisites: Modules 2 and 4–8

Goal: Become comfortable making controlled changes on routers and switches.

Learn in this order

- Packet Tracer device selection, cabling, console access, and Simulation vs Realtime modes.
- CLI navigation: enable, configure terminal, interface, exit/end, show, copy running-config startup-config.
- Hostnames, passwords/secret, console and VTY access, login, SSH management basics.
- Interface descriptions, speed/duplex awareness, no shutdown, IPv4/IPv6 addressing.
- Show commands: show running-config, show interfaces, show ip interface brief, show version.
- Configuration hygiene: comments/descriptions, least privilege, save only verified configurations.
- Safe change process: record baseline, make one change, verify, save, document rollback.

Hands-on lab / evidence

Build a two-switch, two-PC topology. Configure hostnames, management access, interface descriptions, and save the configuration. Verify with show commands.

Move on when you can

You can navigate the CLI, configure an interface, verify status, and save a known-good configuration.

Module 10: VLANs, Trunking, and Inter-VLAN Concepts

Prerequisites: Module 9

Goal: Segment a LAN into separate broadcast domains and understand how traffic crosses between them.

Learn in this order

- VLAN purpose, access ports, VLAN IDs, default VLAN cautions.
- Create/name VLANs; assign access ports; verify VLAN membership.
- 802.1Q trunking, tagged vs untagged traffic, native VLAN concept and mismatch risk.
- Allowed VLAN lists and pruning concept.
- Inter-VLAN routing with router-on-a-stick and Layer 3 switch SVIs.
- Separate user, server, voice, guest, and management networks.
- Common faults: wrong access VLAN, trunk not formed, missing VLAN, wrong gateway, native VLAN mismatch.

Hands-on lab / evidence

Create three VLANs and connect two PCs per VLAN. Verify same-VLAN connectivity, then add inter-VLAN routing and apply a simple traffic policy.

Move on when you can

You can predict whether two hosts should communicate and diagnose the VLAN/trunk/gateway layer when they cannot.

Module 11: Spanning Tree and Layer 2 Resilience

Prerequisites: Module 10

Goal: Prevent switching loops while maintaining redundant links.

Learn in this order

- Why Layer 2 loops are dangerous: broadcast storms, duplicate frames, MAC table instability.
- STP purpose, root bridge, bridge ID, path cost, port roles/states at a conceptual level.
- RSTP and faster convergence; understand common Cisco terminology.
- Root bridge selection and deliberate placement.
- PortFast and BPDU Guard purpose and safe edge-port use.
- EtherChannel/link aggregation concepts and LACP basics.
- Troubleshoot blocked ports, unexpected root bridge, mismatched channel settings.

Hands-on lab / evidence

Create redundant switch links in Packet Tracer. Observe STP blocking a path, select a root bridge deliberately, and bundle compatible links where supported.

Move on when you can

You can explain why a redundant physical link may be logically blocked and how STP prevents loops.

Module 12: Routing Fundamentals and Routing Tables

Prerequisites: Modules 5–9

Goal: Understand how routers select a path before learning routing protocols.

Learn in this order

- Router interfaces, directly connected routes, next hop, exit interface, default route.
- Longest-prefix match; route selection vs administrative distance vs metric.
- Connected, local, static, dynamic, and default routes.
- Routing table fields and how a packet is forwarded hop by hop.
- Control plane vs data plane; routing vs switching.
- ICMP time exceeded and traceroute path interpretation.
- Why a return route is necessary; asymmetric paths at a basic level.

Hands-on lab / evidence

Connect two IP subnets through a router. Inspect the routing table and follow a packet from source subnet to destination subnet.

Move on when you can

You can use a routing table to predict the next hop and explain longest-prefix match.

Module 13: Static Routing, Default Routing, and OSPF

Prerequisites: Module 12

Goal: Configure small routed networks and then introduce a dynamic routing protocol.

Learn in this order

- Configure IPv4 and IPv6 static routes and default routes.
- Recursive vs directly attached next-hop routes; floating static route concept.
- Dynamic routing purpose and tradeoffs; control-plane neighbor relationships.

- OSPF basics: areas, router ID, neighbors, hello/dead timers, cost, LSDB, SPF concept.
- Single-area OSPF configuration and verification; advertised networks/interfaces.
- Route redistribution as a concept only; avoid premature complex redistribution labs.
- Troubleshoot adjacency failures, wrong networks, passive interfaces, duplicate router IDs, and missing routes.

Hands-on lab / evidence

Build a three-router topology. Start with static routes, replace/add OSPF, shut a link, and observe route reconvergence.

Move on when you can

You can diagnose a missing route and explain when static routes versus OSPF make sense.

Module 14: DHCP, DNS, NAT/PAT, and Network Services

Prerequisites: Modules 8, 10, and 12–13

Goal: Connect addressing and routing knowledge to common production network services.

Learn in this order

- DHCP server, scopes, exclusions, reservations, lease duration, relay/IP helper concept.
- DNS zones and records; client resolver settings; forward/reverse lookup troubleshooting.
- NAT purpose; inside/outside terminology; static NAT, dynamic NAT, PAT/overload.
- Private-to-public translation, port translation, state table, return traffic.
- NTP, syslog, SNMP and AAA/RADIUS/TACACS+ roles at an introductory level.
- QoS concepts: classification, marking, queuing, congestion, latency-sensitive voice/video.
- Service dependency troubleshooting: link → IP → gateway/route → DNS → application.

Hands-on lab / evidence

Build a small office topology with DHCP and NAT/PAT in Packet Tracer. Test whether a client can reach an external simulated network by IP and by name.

Move on when you can

You can explain why IP connectivity can work while name resolution fails, and what NAT does and does not do.

Module 15: Access Control Lists and Network Security Basics

Prerequisites: Modules 10, 12, and 14

Goal: Control traffic intentionally and avoid confusing basic filtering with complete security.

Learn in this order

- Threat model basics: attack surface, least privilege, defense in depth, segmentation.
- Standard vs extended IPv4 ACL concepts; order of rules and implicit deny.
- Wildcard masks, source/destination, protocol, port; inbound vs outbound placement.
- IPv6 ACL awareness and management-plane protection.
- Secure device access: SSH rather than Telnet, strong secrets, role separation, disable unused services/ports.
- Port security and DHCP snooping/DAI concepts; understand platform and lab limitations.
- Firewall vs router ACL vs stateful firewall; VPN purpose and encryption basics.
- Backups, patching, change control, incident evidence, and safe handling of credentials.

Hands-on lab / evidence

Apply an ACL that allows only the intended traffic between user and server VLANs. Test both allowed and denied cases and document the rule order.

Move on when you can

You can state precisely which traffic a rule permits/blocks and verify the result rather than assuming it works.

Module 16: Wireless Networking

Prerequisites: Modules 8–10 and 14–15

Goal: Understand how Wi-Fi fits into a switched and routed network.

Learn in this order

- 802.11 overview, 2.4/5/6 GHz bands where supported, channels, channel width, interference.
- SSID, BSSID, access point, wireless client, controller-managed vs standalone AP.
- Signal strength vs throughput; attenuation, roaming, airtime contention.
- WPA2/WPA3, personal vs enterprise authentication, 802.1X/RADIUS concept.
- Guest wireless isolation, VLAN mapping, captive portal concept.
- DHCP/DNS/gateway troubleshooting for wireless clients.
- Survey basics and responsible use of wireless scanning tools.

Hands-on lab / evidence

Design a home/office Wi-Fi plan: placement, channels, security, guest separation, and wired uplinks. If available, compare signal and throughput at multiple spots.

Move on when you can

You can troubleshoot whether a Wi-Fi problem is radio, authentication, DHCP, DNS, or upstream connectivity.

Module 17: Systematic Troubleshooting

Prerequisites: Modules 1–16

Goal: Develop the operational habit employers value: isolate faults efficiently and prove the fix.

Learn in this order

- Define symptoms, scope, recent changes, business impact, and expected behavior.
- Use a layered flow: physical/link → VLAN/MAC → IP/subnet/gateway → route → DNS/service → application.
- Tools: ping, traceroute/tracert, ipconfig/ip, arp, route, nslookup/dig, curl, ss/netstat, Wireshark.
- Read interface counters, logs, routing tables, MAC tables, ARP/neighbor tables, and ACL counters.
- Common faults: wrong mask, duplicate IP, wrong VLAN, bad trunk, missing route, DNS failure, MTU, duplex, ACL/NAT.
- Hypothesis-driven testing: change one variable at a time; compare before/after.
- Write incident notes: impact, timeline, evidence, root cause, fix, prevention, rollback.

Hands-on lab / evidence

Create at least six intentional faults in earlier labs, such as wrong gateway, wrong VLAN, missing route, bad DNS, blocked ACL, and shutdown interface. Diagnose without looking at the answer first.

Move on when you can

You can isolate the failing layer, show evidence for the root cause, and explain why the fix works.

Module 18: Linux Networking and Server Basics

Prerequisites: Modules 2, 8, 12, and 14–17

Goal: Gain practical skill with the systems that commonly sit beside networks: DNS/DHCP services, web servers, firewalls, and monitoring.

Learn in this order

- Linux interface/address/route inspection with ip addr, ip link, ip route, resolvectl.
- Listening sockets and processes: ss -tulpn, systemctl, journalctl.
- Host firewall concepts with ufw/nftables; allow only required services.
- SSH keys, permissions, sudo, updates, and secure remote administration.
- Basic services: web server, DNS client tools, DHCP server concept, file transfer.
- Name resolution order and resolver configuration; local hosts file.
- Virtual machines, snapshots, virtual NICs, bridged/NAT/host-only networks.

Hands-on lab / evidence

Use a Linux VM as a server and a second machine/VM as a client. Host a simple web page, test reachability by IP/name, inspect listening ports, and apply a firewall rule.

Move on when you can

You can separate a Linux service problem from an IP routing or firewall problem.

Module 19: Monitoring, Logs, and Network Documentation

Prerequisites: Modules 9–18

Goal: Learn to operate a network after deployment, not just configure it once.

Learn in this order

- Baseline metrics: availability, latency, loss, interface utilization, errors/discards, CPU/memory.
- SNMP polling and traps; SNMPv3 security concept; syslog levels and centralized logging.
- Flow visibility: NetFlow/IPFIX/sFlow concepts and what traffic records can reveal.
- Monitoring systems and dashboards; alert thresholds, false positives, escalation.
- Configuration backup/versioning; compare diffs; protect secrets in repositories.
- Network diagrams, IP address management (IPAM), port maps, asset inventory, cable labels.
- Runbooks, maintenance windows, change tickets, rollback plans, post-change verification.

Hands-on lab / evidence

Create a topology diagram, IP plan, device inventory, interface/port map, and a troubleshooting runbook for one lab. Capture a baseline and one failure event.

Move on when you can

Another person could understand your topology and repeat your troubleshooting steps from your documentation.

Module 20: Virtual Lab Engineering: Packet Tracer to GNS3/EVE-NG

Prerequisites: Modules 9–19

Goal: Move from guided simulation toward more realistic multi-vendor or multi-OS environments.

Learn in this order

- Packet Tracer strengths and limitations; when its simulated commands differ from real platforms.
- Virtualization concepts: hypervisor, virtual NIC, bridge, NAT network, host-only network.
- GNS3 or EVE-NG architecture, images, resource requirements, and legal image sourcing.
- Connect routers, switches, Linux VMs, firewalls, and client machines in one topology.
- Snapshot/checkpoint discipline and lab reset strategy.
- Resource management: RAM/CPU, lightweight topologies, avoid running everything simultaneously.

- Keep lab files, configs, diagrams, and test plans in version control.

Hands-on lab / evidence

Rebuild one earlier Packet Tracer design with a Linux endpoint and more realistic virtual nodes where your hardware and available legal images permit.

Move on when you can

You understand what the simulator is abstracting and can build, save, reset, and document a reproducible lab.

Module 21: Network Automation and Scripting

Prerequisites: Modules 2, 9, 12–15, 18–20

Goal: Automate repeatable tasks only after manual configuration and verification are understood.

Learn in this order

- Python foundations: variables, strings, lists/dictionaries, loops, functions, files, exceptions.
- JSON, YAML, CSV, regular expressions, structured data and parsing command output.
- Git fundamentals: commits, branches, diffs, README files, secrets hygiene.
- APIs and HTTP basics; authentication, tokens, status codes, pagination concept.
- SSH automation concepts and libraries such as Netmiko; do not run scripts against networks without authorization.
- Configuration templates and data-driven device inventories.
- Ansible fundamentals: inventory, playbooks, variables, idempotence, check/diff modes.
- Backups, configuration drift checks, validation, logging, dry-run and rollback planning.

Hands-on lab / evidence

Write a Python script that validates an IP plan or parses a saved command output. Then create a small authorized lab script that collects device facts or backs up configurations.

Move on when you can

Your script handles errors, avoids hard-coded secrets, logs what it did, and verifies results instead of blindly pushing changes.

Module 22: High Availability, Design, and Performance

Prerequisites: Modules 10–19

Goal: Design networks for failure, growth, and predictable operation rather than only making a single ping succeed.

Learn in this order

- Requirements gathering: users, applications, traffic, security, growth, budget, support model.
- Hierarchical campus design: access, distribution, core; collapsed core for smaller sites.
- Redundancy, single points of failure, convergence, maintenance and failure domains.
- First-hop redundancy concepts (HSRP/VRRP), link aggregation, routing failover.
- QoS policy design basics; voice/video, queuing and congestion handling.
- MTU/MSS, fragmentation, path MTU discovery and symptoms of size-related failures.
- Capacity planning, oversubscription, latency budgets, availability targets and tradeoffs.
- IPv4/IPv6 addressing plans, summarization, management plane, out-of-band access.

Hands-on lab / evidence

Design a two-floor office network with user, server, guest, and management segments. Include an IP plan, redundancy choices, security boundaries, diagram, and failure tests.

Move on when you can

You can justify design choices, identify single points of failure, and describe expected behavior when a link or device fails.

Module 23: ISP and Service-Provider Networking

Prerequisites: Modules 6–8, 12–14, 17–22

Goal: Build on enterprise routing to understand how internet providers deliver connectivity.

Learn in this order

- Access networks: fiber-to-the-home, GPON/XGS-PON concepts, ONT/ONU, OLT, splitter, optical power budget awareness.
- Last-mile technologies: Ethernet, fiber, DOCSIS, wireless access; broadband handoff and customer-premises equipment.
- PPPoE, VLAN-based subscriber access, DHCP/IPoE, RADIUS/AAA concepts.
- CGNAT, public vs private WAN addresses, IPv4 scarcity, IPv6 delegation concepts.
- BGP purpose, autonomous systems, eBGP/iBGP, prefix announcements, path selection at a conceptual level.
- Peering, transit, IXPs, upstream providers, route filtering and prefix hygiene.
- MPLS and L2/L3 VPN concepts; understand purpose before detailed configuration.
- ISP operations: NOC workflow, ticket triage, outage scope, latency/loss checks, escalation, maintenance communication.
- Customer-side troubleshooting: optical/link status, router WAN address, PPPoE credentials, VLAN tagging, DNS, Wi-Fi.

Hands-on lab / evidence

Create a simulated ISP with customer edge, provider edge, and upstream network. Document subscriber addressing, NAT/IPv6 plan, and a simple multi-AS BGP lab only after static routing and OSPF are comfortable.

Move on when you can

You can explain the path from customer ONT/router through an ISP to upstream transit and distinguish home-router problems from provider-side faults.

Module 24: Portfolio, Certification Readiness, and Job Preparation

Prerequisites: All previous modules

Goal: Turn study into evidence that can be evaluated by a hiring manager or mentor.

Learn in this order

- Use CCNA 200-301 objectives as a checklist and map each objective to a lab or explanation.
- Practice subnetting, CLI output interpretation, configuration, troubleshooting, and scenario questions.
- Create a GitHub repository with clear folders, README, topology diagrams, sanitized configs, lab steps, and lessons learned.
- For each lab: goal, prerequisites, topology, addressing table, configuration, verification, failure injection, fix, screenshots/pcaps, conclusion.
- Portfolio projects: segmented small-office network; routed multi-site network with OSPF; ACL/security segmentation; Linux server plus monitoring; ISP-style lab.
- Prepare to explain tradeoffs and troubleshooting process, not only recite definitions.
- Resume evidence: projects, tools, technical writeups, teamwork, and honest skill levels.
- Certification is a checkpoint, not a substitute for labs. Check current exam objectives, pricing, and policies before paying.

Hands-on lab / evidence

Publish three polished projects and at least six troubleshooting case studies. Ensure no passwords, private keys, personal data, or unlicensed device images are committed.

Move on when you can

You can demo your topology, reproduce it, explain every major choice, and troubleshoot a deliberate fault live.

Practical lab and GitHub workflow

Use the same repeatable format for every meaningful experiment. This turns daily practice into a portfolio rather than a pile of disconnected notes.

- **1. Objective:** What exact behavior are you trying to make work or understand?
- **2. Prerequisites:** List the module concepts and commands needed before starting.
- **3. Topology:** Add a diagram, device names, links, VLANs, and IP addressing table.
- **4. Baseline:** Record the initial state and expected result.
- **5. Configure:** Make small, explainable changes. Save the relevant commands/configuration.
- **6. Verify:** Use show commands, pings, route tables, packet captures, and service tests.
- **7. Break it intentionally:** Change one setting to create a controlled failure.
- **8. Diagnose and fix:** Record evidence, root cause, correction, and retest results.
- **9. Reflect:** State what surprised you, what limitation the tool had, and what you would do in production.
- **10. Commit:** Use a specific Git commit message such as “Lab 04: verify ARP and switch MAC learning”.

Suggested repository structure

```
network-engineering-journey/
README.md
01-foundations/
02-cli-and-os/
03-protocol-models/
04-ethernet-arp-switching/
05-ipv4-and-binary/
06-subnetting/
07-ipv6/
08-core-protocols/
09-device-cli/
10-vlans-and-trunks/
11-stp-and-etherchannel/
12-routing/
13-ospf/
14-network-services/
15-acls-and-security/
16-wireless/
17-troubleshooting-cases/
18-linux-networking/
19-monitoring-and-docs/
20-virtual-labs/
21-automation/
22-design-projects/
23-isp-labs/
24-capstone-projects/
```

Do not upload credentials, private keys, real customer data, or proprietary/unlicensed images. Sanitize configurations before publishing.

Recommended tools by stage

When	Tools	Use
Start now	Cisco Packet Tracer	Build LANs, VLANs, routing, ACL, NAT and OSPF labs without physical hardware.
Start now	Wireshark	Inspect frames, ARP, DNS, TCP handshakes, ICMP and retransmissions.
Start now	Windows Terminal / Linux VM	Practice real commands and understand host-side networking.
Start now	GitHub + an editor	Version-control lab notes, diagrams, sanitized configs and results.
After fundamentals	GNS3 or EVE-NG	Build more realistic multi-node labs; hardware/resource needs vary.
After routing basics	Python, Netmiko, Ansible	Automate repeatable, authorized lab tasks and configuration checks.
After monitoring basics	SNMP/syslog dashboard tools	Observe availability, interface counters, logs and alerts.

Learning resources: Use Jeremy's IT Lab as a structured CCNA course, Cisco's official CCNA exam topics as a coverage checklist, Practical Networking for clear visual explanations, and NetworkChuck as supplementary motivation/exploration—not as your only curriculum. Verify current exam objectives and certification prices directly with Cisco before making a purchase.

Milestones (not a day-by-day schedule)

- **Foundation milestone:** You can explain packet delivery and use basic host diagnostic commands.
- **LAN milestone:** You can build VLANs/trunks, explain STP, and troubleshoot Layer 2 failures.
- **Routing milestone:** You can subnet independently, configure static routes and OSPF, and interpret route tables.
- **Services/security milestone:** You can configure or simulate DHCP, NAT, ACLs, and basic segmentation.
- **Operations milestone:** You can diagnose intentionally broken networks and write useful incident records.
- **Portfolio milestone:** You have reproducible projects, diagrams, sanitized configs, and troubleshooting evidence.
- **Specialization milestone:** Choose a deeper focus—enterprise networking, ISP/service provider, network automation, or network security—based on which labs you enjoy and perform well in.

Important reality check: Simulators are enough to learn many core concepts at the start, but they do not perfectly reproduce every vendor feature, optical fault, radio condition, performance issue, or production incident. Virtual labs plus packet captures and Linux practice provide strong foundations; real operational experience remains valuable.